

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1976

Flagami 138 kV
August 27, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1976. The Interconnection Customer proposes a 157.7 MW battery energy storage facility interconnecting at the Flagami 138 kV substation in the Kentucky area. The request was received on January 06, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 543 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,277,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Prineville / Wetin-Elgin 138 kV, the Pg&E; Pump / Alameda 138 kV loads to 130.8% of its 262 MVA rating. Project DFAX: 100.0%.

The Prineville Substation / Flagami Substation 138 kV is overloaded to 129.0% of its 259 MVA emergency rating for the loss of the Abajo / Fort Rucker 138 kV. The project contributes 100.0% DFAX (157.7 MW) to this overload.

The Jacinto / PgE Pump 138 kV is overloaded to 128.9% of its 275 MVA emergency rating for the loss of the Prineville / Westin-Elgin 138 kV. The project contributes 100.0% DFAX (157.7 MW) to this overload.

Sheffield / Jacinto 138 kV Line: 128.0% loading (282 MVA) following the outage of the Prineville / Westin-Elgin 138 kV L; DFAX = 100.0%.

The Hill Country 345 / Abajo Substation 138 kV is overloaded to 126.8% of its 289 MVA emergency rating for the loss of the Prineville / Westin-Elgin 138 kV. The project contributes 100.0% DFAX (157.7 MW) to this overload.

The Westin-Elgin / Prineville 138 kV is overloaded to 114.4% of its 326 MVA emergency rating for the loss of the Abajo / Fort Rucker 138 kV. The project contributes 100.0% DFAX (157.7 MW) to this overload.

For the loss of Spatsum Lornex / Latah Jct 138 kV L, the E Broad / Harvest Power Orlando 138 kV loads to 111.2% of its 218 MVA rating. Project DFAX: 28.8%.

The Welfare / Tomahawk-Atc Substation 138 kV is overloaded to 108.6% of its 213 MVA emergency rating for the loss of the Spatsum Lornex / Latah Junction 138 kV. The project contributes 26.1% DFAX (41.1 MW) to this overload.

The Spry / Alamea 138 kV is overloaded to 105.3% of its 229 MVA emergency rating for the loss of the Westin-Elgin / Prineville 138 kV. The project contributes 100.0% DFAX (157.7 MW) to this overload.

Harvest Power Orlando / Frontier 138 kV: 103.5% loading (410 MVA) following the outage of the Spatsum Lornex / Latah Jct 138 kV; DFAX = 44.8%.

Frontier / Delano 138 kV: 101.8% loading (202 MVA) following the outage of the Spatsum Lornex / McClellan Ar Substation 138 kV L; DFAX = 24.9%.

Sheffield / Curry County 138 kV L: 101.6% loading (290 MVA) following the outage of the Westin-Elgin / Prineville 138 kV; DFAX = 100.0%.

Rec Bradford / Curry County 138 kV: 101.5% loading (296 MVA) following the outage of the Prineville / Westin-Elgin 138 kV; DFAX = 100.0%.

The Rec Bradford / Potato Rapids 138 kV is overloaded to 100.9% of its 316 MVA emergency rating for the loss of the Prineville / Westin-Elgin 138 kV. The project contributes 100.0% DFAX (157.7 MW) to this overload.

For the loss of Latah Jct / Spatsum Lornex 138 kV, the Kettle / Country Club 2 Substation 138 kV Line loads to 100.9% of its 299 MVA rating. Project DFAX: 27.6%.

For the loss of Spatsum Lornex Substation / Latah Jct 138 kV, the Country Club 2 / Spatsum Lornex Substation 138 kV loads to 100.6% of its 306 MVA rating. Project DFAX: 27.6%.

The Osawatomie / Sunset N 138 kV is overloaded to 100.3% of its 241 MVA emergency rating for the loss of the Wagner Lake / Osawatomie 230 kV. The project contributes 9.3% DFAX (14.6 MW) to this overload.

For the loss of Westin-Elgin / Prineville 138 kV, the Peacock / Potato Rapids 138 kV line loads to 100.3% of its 344 MVA rating. Project DFAX: 100.0%.

The Peacock / Hill Country 345 138 kV is overloaded to 100.2% of its 351 MVA emergency rating for the loss of the Prineville / Westin-Elgin 138 kV. The project contributes 100.0% DFAX (157.7 MW) to this overload.

S / Cortez 138 kV: 100.0% loading (197 MVA) following the outage of the Summit Lake Wps / Edna 138 kV; DFAX = 9.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alameda / Pg&E; Pump Substation 138 kV	Rebuild line with higher-capacity conductor	\$7,239,000	100% (\$7,239,000)
Prineville Substation / Flagami Substation 138 kV	Rebuild line with higher-capacity conductor	\$4,273,000	68% (\$2,910,000)
Jacinto / PgE Pump 138 kV	Rebuild line with higher-capacity conductor	\$6,065,000	100% (\$6,065,000)
Sheffield / Jacinto 138 kV Line	Rebuild line with higher-capacity conductor	\$3,959,000	100% (\$3,959,000)
Hill Counry 345 / Abajo Substation 138 kV L	Rebuild line with higher-capacity conductor	\$3,672,000	68% (\$2,500,000)
Westin-Elgin / Prineville 138 kV	Rebuild line with higher-capacity conductor	\$18,824,000	68% (\$12,818,000)
Harvest Power Orlando / East Broad 138 kV	Rebuild line with higher-capacity conductor	\$6,600,000	24% (\$1,569,000)
Welfare / Tomahawk-Atc Substation 138 kV	Reconductor line	\$3,652,000	24% (\$892,000)
Alameda / Spry 138 kV	Reconductor line	\$3,904,000	100% (\$3,904,000)
Harvest Power Orlando / Frontier 138 kV	Reconductor line	\$9,162,000	24% (\$2,191,000)
Frontier / Delano 138 kV	Reconductor line	\$1,808,000	30% (\$536,000)
Sheffield / Curry County 138 kV L	Reconductor line	\$3,458,000	68% (\$2,355,000)
Rec Bradford / Curry County 138 kV	Reconductor line	\$2,356,000	68% (\$1,604,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Potato Rapids Substation / Rec Bradford 138 kV	Reconductor line	\$2,286,000	68% (\$1,557,000)
Kettle / Country Club 2 Substation 138 kV Line	Reconductor line	\$3,975,000	22% (\$857,000)
Spatsum Lornex / Country Club 2 138 kV	Reconductor line	\$4,178,000	22% (\$900,000)
Osawatomie / Sunset N 138 kV	Reconductor line	\$539,000	4% (\$19,000)
Peacock / Potato Rapids 138 kV line	Reconductor line	\$4,125,000	68% (\$2,808,000)
Peacock / Hill Country 345 138 kV	Reconductor line	\$1,880,000	68% (\$1,280,000)
S / Cortez 138 kV	Reconductor line	\$2,633,000	12% (\$314,000)
	Total allocated to project		\$56,277,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.